Sensor applications

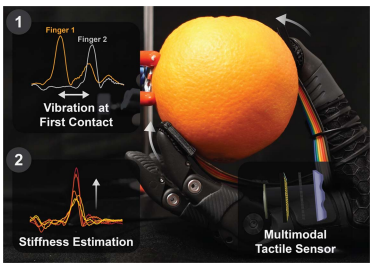
Vibrations at First Contact Encode Object Stiffness Before Grasp Completion

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Abstract—Current methods for stiffness estimation in artificial dexterous systems depend on force and displacement measurements after an object is fully grasped. This delays grasp adjustments until the grasping process is complete. The proposed approach in this study introduces a novel method for stiffness estimation at first contact, exploiting the time window created by "asymmetry" in grasp dynamics when one finger contacts an object before the others. We integrated a biomimetic tactile sensor inspired by human skin anatomy into an artificial dexterous hand that simultaneously captures both force and vibration information. Vibrational signals from the first finger's initial contact were found to encode stiffness information, achieving high-accuracy discrimination (98.6%) and regression (error: 2.39 Shore A) of object stiffness using machine learning models. Despite the brief temporal window (mean: 16.65 ms), models could infer stiffness in under 1.5 ms, allowing time for grasp modulation with the remaining fingers before the object is fully secured. Vibration sensing in object grasping establishes a new paradigm for rapid stiffness estimation, with significant potential for advancing dexterous manipulation in delicate tasks for robotics.



Index Terms—Sensor applications, adaptive control, grasp modulation, prosthetics, stiffness estimation, tactile sensing.

I. INTRODUCTION

In object manipulation, tactile feedback is vital for adjusting grip to the mechanical properties of contacted objects. Among these, stiffness perception is essential in ensuring safe and effective interaction, particularly in delicate tasks involving compliant or brittle objects [1]. Introducing this capability in artificial dexterous hands remains a significant challenge, particularly for precise stiffness estimation [2]. Traditional approaches for stiffness estimation predominantly rely on tactile force sensors or measuring finger displacements under applied forces during grasps [3], [4], [5], [6]. In these methods, contact stiffness is typically inferred from the fingers' motor current and angular displacement or the ratio of force to deformation. These approaches become effective only after complete contact has been established [6], which increases the risk of damaging brittle objects during delicate tasks. Therefore, methods that estimate stiffness rapidly before the grasp is complete are crucial for improving dexterous manipulation.

This study introduces a novel method for estimating stiffness at first contact using vibration sensing, leveraging the previously unexamined brief temporal asymmetry between the first and second finger contacts during a grasp. We demonstrate this approach of utilizing multimodal tactile sensing—capturing both vibration and force—with a dexterous hand performing pinch grasps, a ubiquitous action in daily manipulation tasks [7]. To demonstrate that vibration sensing encodes object stiffness during the crucial window where force data cannot, we collected data from silicone blocks with known stiffness to train machine learning models for contact detection, stiffness discrimination, and regression within milliseconds of contact. We then validated the stiffness estimation on real-world objects.

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II. BIOMIMETIC TACTILE SENSOR DESIGN

We developed a biomimetic multilayer tactile sensor to capture force and vibration information, mimicking the mechanoreceptor distribution in human skin [8], as illustrated in Fig. 1. Merkel discs and Meissner corpuscles near the skin surface detect sustained pressure and light touch, while deeper Pacinian corpuscles sense high-frequency vibrations, crucial for transient contact events. By emulating both these functions, the designed sensor achieves multimodal tactile sensing.

We integrated this sensor into the distal phalanges of the index finger and thumb on a dexterous hand, the TASKA HandGen2 (TASKA Prosthetics, New Zealand). Each sensor assembly features grip tips with a 3 mm-thick molded silicone layer of 65 Shore A stiffness. The layer has dimpled extrusions on its underside, directing applied forces precisely onto the underlying force sensors. Beneath the silicone layer, a 20 mm × 15 mm printed circuit board (PCB) is designed to house a 3 × 2 piezoresistive force sensor array on its front, mimicking the superficial placement of Merkel discs and Meissner corpuscles. The force sensor array features interwoven traces serving as the two electrodes for the piezoresistive Velostat material layered on top. This 0.1 mm-thick sheet has a volume resistivity of $< 500\,\Omega$-cm and a surface resistivity of $< 31\,000\ \Omega/\text{cm}^2$. For vibration sensing, a 10 mm diameter lead zirconate titanate (PZT) piezoelectric element is mounted on the reverse side of the PCB, emulating the deeper placement of Pacinian corpuscles responsible for encoding transient vibration information. The assembly is encased in a custom 3-D-printed

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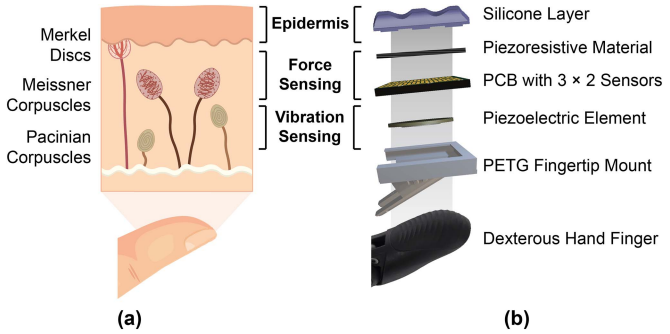


Fig. 1. (a) Mechanoreceptor placement in human skin. (b) Corresponding biomimetic tactile sensor for vibration and force sensing.

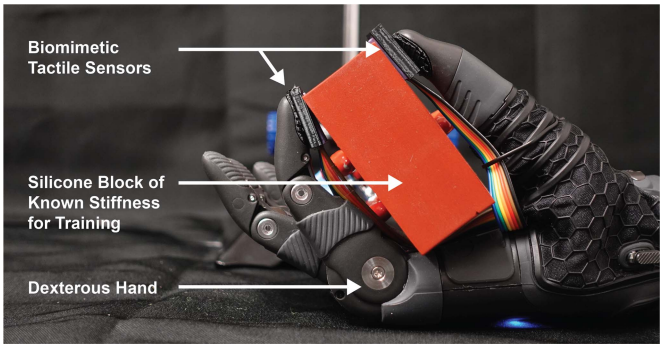


Fig. 2. Training data collection with a dexterous hand equipped with biomimetic tactile sensors, performing pinch tasks on silicone blocks.

polyethylene terephthalate glycol fingertip mount, custom designed for the hand and secured with screws.

The vibration and force sensor outputs are buffered and amplified using an analog circuit for signal processing. Data acquisition is facilitated by a 10-bit resolution analog-to-digital converter integrated into a Teensy 4.0 MCU (ARM Cortex-M7, NXP iMXRT1062) operating at a clock speed of 600 MHz. Signals are recorded at an average sampling rate of 4936 Hz. The six piezoresistive sensor signals are time-division multiplexed using a row-column addressing scheme and measured via a voltage divider circuit. The analog output from the piezoelectric element is buffered through a noninverting amplifier and offset to 1.65 V, enabling the detection of both positive and negative spikes within a 3.3 V range. Once buffered into an array, the data is transmitted to a connected computer over the serial port, managed by a Python 3.8 script for further processing.

The vibration data undergoes exponential smoothing ($\alpha = 0.5$) in real time. Force data is smoothed using a sliding window average. After collecting the data, a Savitzky–Golay filter eliminates high-frequency noise without distorting key vibrational patterns.

III. STIFFNESS ESTIMATION AT FIRST CONTACT

A. Training Data Collection

The experimental setup for stiffness estimation involved pinching silicone blocks to collect training data. Five blocks with dimensions of 7 cm ×7 cm× 3.5 cm were fabricated, representing stiffness values of 10, 20, 29, 43, and 60 Shore A, verified with a PCE-DD-A durometer (PCE Instruments, Germany). Desired stiffness levels were achieved by proportionally blending Smooth-On Mold Max 60 and Mold Max 10T (Smooth-On Inc., USA) silicone compounds.

The dexterous hand is positioned palm-up on a table, with silicone blocks suspended above in its grasp using a vertical clamp, as shown in Fig. 2. In total, 500 pinches per stiffness level (2500 total) were performed to train the models, with slight adjustments to block placement to introduce natural variations and enhance dataset robustness.

B. Contact Detection

To detect the moment of contact from vibration data, we established ground truth using thin copper strips applied to the finger and object surface, connected to the MCU's digital input pin to register contact events. This timing was independently validated with LEDs connected to the contact strips and recorded at 960 fps using a Sony RX100 V camera (Sony Inc., Japan). We then implemented two approaches for contact detection using this ground truth data: a thresholding method for its simplicity and computational efficiency, and a machine learning-based method using a support vector machine (SVM) with a radial basis function kernel, chosen for its ability to handle nonlinear relationships in the data [9]. The thresholding method identified contact by applying a fixed threshold at three times the standard deviation (3σ) of steady-state signals [10]. Both approaches operated on 20 ms sliding windows, consisting of 17 ms of preceding data and 3 ms of newly acquired data.

C. Stiffness Discrimination and Regression Models

To evaluate whether vibrational information at first contact encodes object stiffness, we implemented two approaches for extracting this data: another SVM and a convolutional neural network (CNN), selected for its ability to learn spatiotemporal representations from high-dimensional tactile input [11], [12], [13]. Both models were trained in discrimination and regression contexts to estimate stiffness from 2500 silicone block pinches using a 90–10 training-validation split. The maximum amplitude was used as a proxy metric for transient mechanical shock, reflecting synchronized piezoelectric activation under transient impulses, consistent with wave summation in PZT sensors [14].

An SVM was trained on 15 ms windows of vibrational data captured immediately after contact, leveraging its ability to model nonlinear relationships between vibrational features and stiffness levels. Hyperparameter optimization was performed to fine-tune kernel parameters, maximizing discrimination accuracy and regression performance while maintaining computational efficiency.

The CNN approach utilized the *EfficientNetV2* architecture, chosen for its efficiency in capturing complex data patterns [15]. Inputs consisted of 15 ms temporal windows immediately after contact, identical in size to those used for the SVM. The training was performed over 40 epochs using the Adam optimizer with an initial learning rate of 0.001 and the mean squared error (MSE) loss function. A step learning rate scheduler reduced the learning rate by half every five epochs to ensure stability and convergence.

D. Model Validation and Generalization

To assess the generalizability of stiffness estimation learned on silicone blocks, we tested the models on objects representing real-world scenarios where a dexterous hand must interact with materials of varying stiffness. The selected objects included two apples (28, 26 Shore A), two oranges (35, 37 Shore A), two tennis balls (45, 46 Shore A), and two avocados (59, 67 Shore A). This selection covered a range of stiffness levels and material properties, reflecting the variability of everyday tasks and the challenges of adaptive manipulation. These

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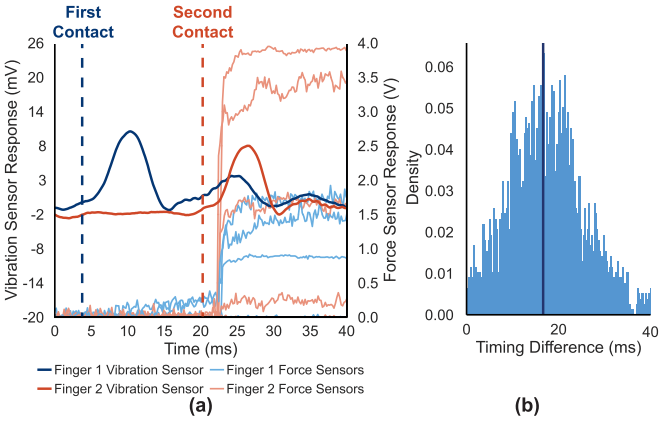


Fig. 3. (a) Timing of first and second finger contacts with corresponding force and vibration sensor outputs. (b) Distribution of time intervals between the first and second finger contacts.

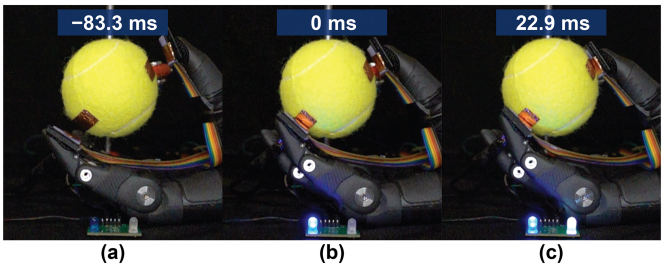


Fig. 4. Contact validation using high-speed photography: (a) Frame before contact, (b) frame at first contact, and (c) frame at second contact.

objects were tested exclusively in a regression context to simulate real-world encounters with unknown objects rather than evaluating similarities with precharacterized ones through discrimination. Vibrational data from the initial 15 ms after contact served as input for the SVM and CNN models, which predicted stiffness values on a continuous scale. The model outputs were evaluated against ground truth and measured with the same Shore A durometer used for the training blocks using an MSE metric.

IV. RESULTS AND DISCUSSION

For contact detection, the thresholding method achieved 90.6% accuracy, as calculated from the confusion matrix, but was hindered by false positives. In contrast, the SVM achieved a perfect 100% accuracy. Despite the computational efficiency of thresholding, its limitations led to the selection of the SVM for subsequent tasks.

The analysis of the pinch grasp event, where both fingers sequentially contact the object, reveals a measurable timing discrepancy between the two fingers. This *asymmetry in finger contact*, confirmed by ground truth data from electrical contact strips, as seen in Fig. 3(a), stems from the inherent unevenness of grasp dynamics, with one finger reaching the object before the other [16], [17]. This stochastic phenomenon was quantified for our dataset, yielding a timing distribution with a mean offset of 16.65 ms and a standard deviation of 10.35 ms, as shown in Fig. 3(b). This timing difference was further validated through high-speed photography, as shown in Fig. 4. While human pinch grasps show ∼50 ms interfinger delays [18], the 16.65 ms mean delay in our robotic system is due to faster actuation and shorter initial finger–object distances. During the initial contact by the first

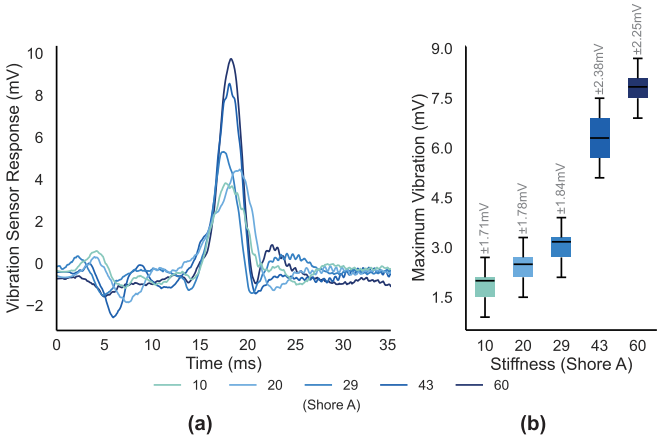


Fig. 5. (a) Representative piezoelectric sensor responses for varying stiffness levels and (b) maximum piezoelectric responses corresponding to different stiffness. Each box represents 500 pinches.

finger, a pronounced vibrational signal is recorded, while force data remains absent due to the lack of a reaction force until the second finger establishes contact, seen in Fig. 3(a). This temporal gap, where one finger contacts the object before the other, provides a critical window to capture vibrational information before force is registered, enabling early data processing for real-time adaptive grasp modulation.

The alignment of contact events across all stiffness blocks revealed an emergent relationship between the piezoelectric disk's response and the stiffness of the block, as seen in Fig. 5(a). Specifically, the amplitude of the piezoelectric response exhibited a proportionality to the stiffness, as seen in Fig. 5(b). This suggests that the amplitude of the piezoelectric signal can reliably represent stiffness.

The SVM approach achieved an overall accuracy of 97.2% on the discrimination task, while the CNN demonstrated a slightly higher accuracy of 98.6%. While both models proved highly effective, the CNN's superior performance stems from its ability to learn and extract complex, high-dimensional features, capturing subtle vibrational patterns associated with stiffness. The SVM demonstrates superior computational efficiency, with an average inference time of less than 1 ms compared to the CNN's 1.4 ms. Both inference times are significantly shorter than the mean time between contacts (16.65 ms), making either approach viable for real-time stiffness estimation before fully grasping the object—a critical advantage given the stochastic nature of grasp dynamics.

Both SVM and CNN approaches demonstrated strong generalization capabilities for estimating stiffness in objects with varying stiffness levels in real-world settings. The models achieved overall MSE values of 4.17 Shore A (< 1 ms inference time) and 2.39 Shore A (< 1.5 ms inference time), respectively, across all objects, as shown in Fig. 6. These accuracies and similar latencies demonstrate that first-contact vibrational sensing can generalize to unseen objects within a range of stiffness comparable to the training data. While inference times were consistently low for both models, they exhibited minimal per-trial variation.

We analyzed regression accuracy over time and found that both models improved rapidly in the first 10 ms, then plateaued. The CNN reached sub-3 Shore A MSE by 8 ms, while the SVM plateaued around 13 ms. This latency–accuracy tradeoff, shown in Fig. 7, highlights the early informativeness of vibrational signals.

The proposed method estimates stiffness at first contact using vibrations, whereas existing force-based approaches require a fully

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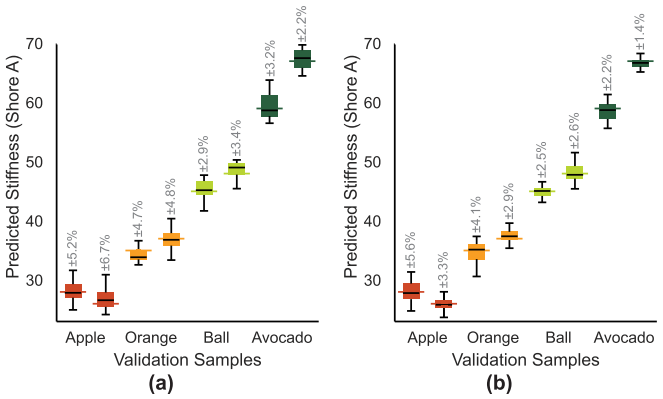


Fig. 6. Stiffness regression on real-world objects with varying stiffness levels (two per group, 20 pinches each) using (a) SVM and (b) CNN models, with true values shown as color bars alongside the box plots.

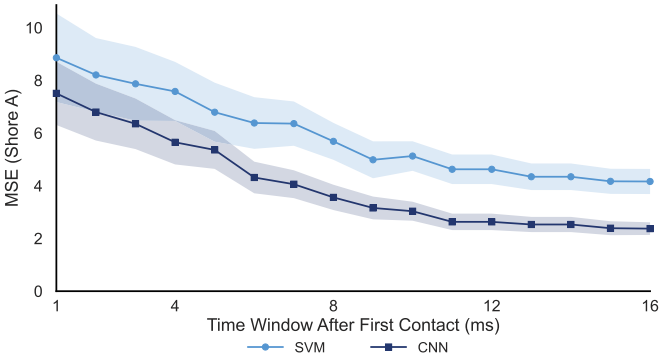


Fig. 7. MSE for stiffness estimation decreases with increasing post-contact window size. Shaded regions represent ±1 standard deviation across trials for SVM and CNN models.

secured grasp before predictions can be made. This opens a critical early window for grasp modulation by the remaining fingers. Although the models are not explicitly trained on asymmetric contact features, the method's constrained sampling window captures vibrational data exclusively from the leading finger, enabling inference prior to full contact. Despite the controlled setup, the method generalizes well, as demonstrated by accurate regression on diverse real-world objects.

Expanding the dataset to include a broader range of objects and interaction scenarios will further enhance the robustness of the models. Optimizing the models for piezoelectric signal characteristics could further enhance performance. Future work will also assess performance on stiffer or irregularly shaped objects, where contact dynamics may differ. Future work should focus on integrating the proposed methods directly into dexterous hand control systems, enabling true real-time grasp modulation by minimizing communication latencies between the sensors and actuators. Integrating this stiffness estimation method into prosthetic hands could enhance safety and confidence in handling delicate objects, improving functionality and quality of life for upper-limb amputees. More broadly, it holds promise for adaptive manipulation in industrial automation, healthcare, and teleoperation.

V. CONCLUSION

This study demonstrates the effectiveness of vibration sensing for real-time stiffness estimation during the critical first-contact phase of manipulation tasks. Vibrational data captured within 15 ms of the first finger's contact enabled accurate stiffness discrimination and regression, with inference times of < 1 ms for the SVM and < 1.5 ms for the CNN. These delays, significantly shorter than the 16.65 ms mean intercontact interval, allow stiffness estimation before the object is fully grasped, providing a temporal advantage over prior methods that rely on complete grasp. The relationship between the piezoelectric response amplitude and stiffness validates vibrational data as a reliable stiffness indicator. Integrating this method could enhance real-time adaptive grasp modulation for delicate objects.

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